COMMERCIAL FIELD BRIEF

Making money off the grid.

How to price the platform — **with** a bundled off-grid device, and **without** one. The moat, the model, the numbers, and the one risk that decides whether "free device" is genius or a money pit.

Prepared by the **Commercial & Design Desk**
Strategy & unit economics · connected-hardware subscriptions

For the Founder
July 2026 · v2

01

THE THESIS

The app is the hook. The system is the business.

Anyone can build a beautiful trail app in a weekend now. What they cannot build in a weekend is a fleet of devices in the desert, a mesh that gets denser every trip, and a library of real recoveries that no competitor has. That is the company. The app just gets people in the door.

The wedge is safety, not maps. Overlanders in the Empty Quarter don't lose sleep over which trail is prettiest — they worry about losing a vehicle in a convoy where there is no signal for 200 km. That fear is specific, acute, and unserved. We lead with **keeping the group together off-grid** and **getting un-stuck** — everything else is supporting cast.

The bet. Give the group a cheap LoRa mesh device so their phones can see each other with no cell signal, wrap it in polished maps and a recovery co-pilot, and charge a subscription that quietly pays for the hardware. WHOOP proved people will happily take a "free" device if the service is worth the monthly.

WHY THIS IS DEFENSIBLE

Hardware	Capital, firmware & certification a cloner can't skip
Deployed fleet	Mesh only works where devices already are — first mover compounds
Data	Every trip and recovery makes the product smarter
Community	Clubs and convoys create switching cost & loyalty

THE POINT

Copy the screens all you like — you can't copy a mesh you never deployed or recoveries you never logged.

02

DEFENSIBILITY

The moat stack

Read it bottom to top. Each layer is harder to copy than the one below — and the value migrates upward as the business matures. If we only ever ship the bottom layer, we're a feature. The plan is to climb.

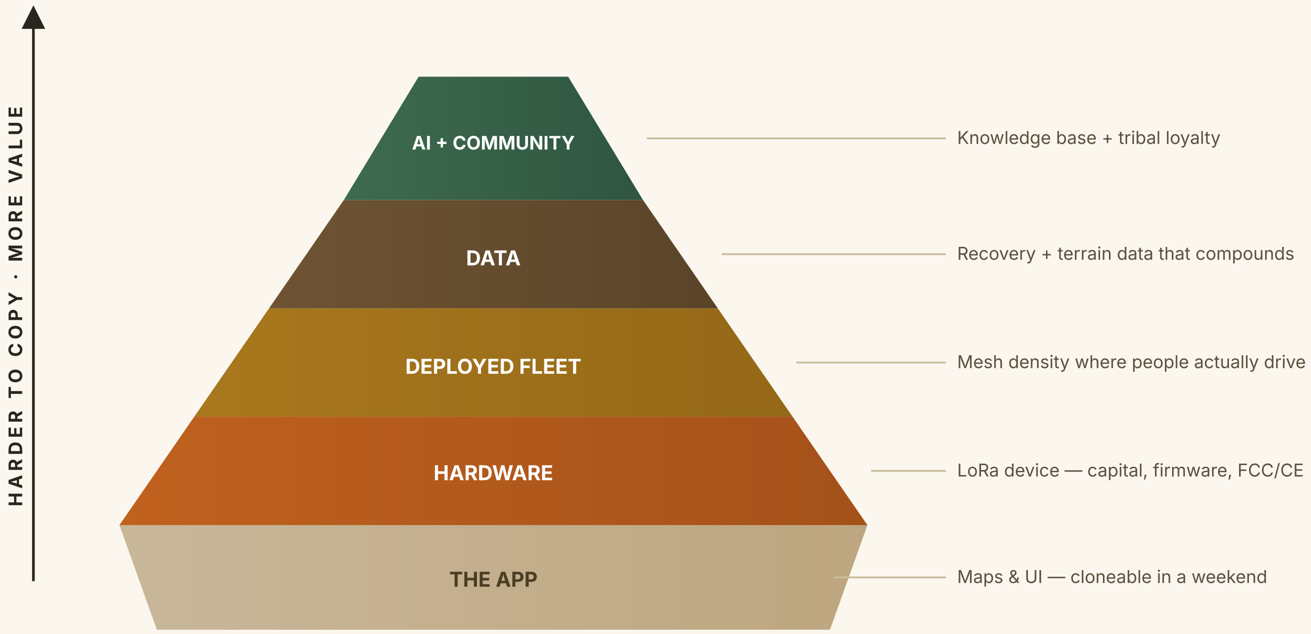


Fig. 1 — Layers of defensibility. The base is exposed to copycats; the apex is nearly impossible to replicate without years of deployment and data.

03

WHERE WE PLAY

Nobody owns the off-grid convoy.

The map apps are built for one person with signal. The satellite gadgets are built for one person in an emergency. Real overland groups off-grid fall straight through the gap — and that gap is the whole opportunity.

CATEGORY	WHO	WHAT THEY NAIL	WHERE THEY LEAVE THE GAP
Trail & map apps	AllTrails, onX Offroad, Gaia GPS	Gorgeous maps, huge trail libraries, cheap subs	Single-user, need signal, no hardware, no group cohesion, no recovery help
Satellite safety	Garmin inReach, Zoleo, SPOT	True global SOS & messaging anywhere	$150–400 hardware + pricey plans; built for solo emergencies, clunky, not live convoy tracking
DIY mesh	Meshtastic community	The LoRa tech works & is cheap	Hobbyist, no maps, no UX, no support, no community product
Overland us	This platform	Polished maps + real-time off-grid group mesh + recovery co-pilot, one accessible bundle	—

We're not competing on **prettier maps***. We're the only one that keeps a* **whole convoy together** *where there is no signal — and helps them* **get un-stuck** *when it goes wrong.*

04

THE BUSINESS MODEL

Subscriptions pay for the "free" device.

The subscription isn't just software margin — it's the engine that funds the hardware, which drives adoption, which builds the moat, which lowers churn, which funds more hardware. Get the loop spinning and it compounds.

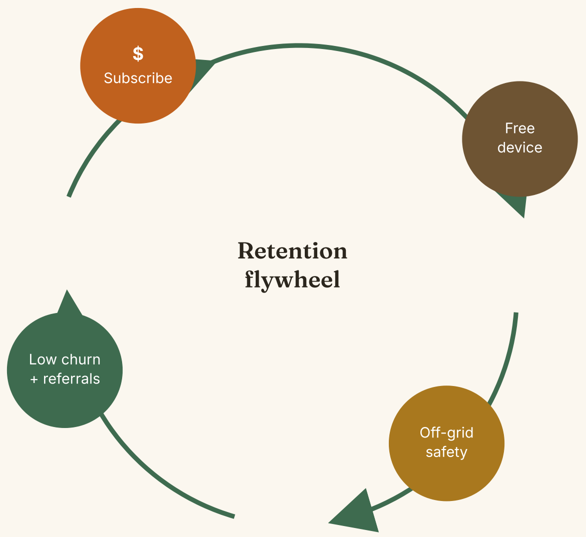


Fig. 2 — Every turn adds members, mesh density and data, which deepens the moat and lowers churn.

HOW EACH NODE PAYS FOR THE NEXT

1	Members subscribe — **~$15/mo on an annual commit**.
2	That funds the **free LoRa device**, offline maps and the recovery co-pilot.
3	Off-grid safety they can't get elsewhere gets used **every single trip**.
4	Daily value = **low churn + referrals** → denser mesh, more data, deeper moat → more members.

> **WHY ANNUAL, NOT MONTHLY**
>
> The 12-month commitment is not a pricing trick — it is the single mechanism that keeps a member subscribed long enough to pay back their device. It is the load-bearing wall of the whole model.

05

UNIT ECONOMICS

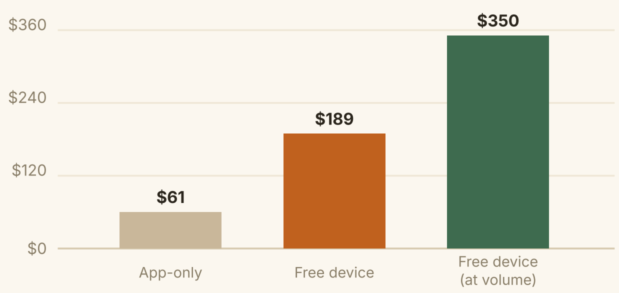
With the device vs. without it.

Four ways to charge. App-only is cheap and fast but shallow. Selling the device protects cash but throttles growth. The free-device model earns 3× the lifetime value and builds the moat — if churn behaves (Section 06).

MODEL	USER UPFRONT	MONTHLY	DEVICE COST	CONTRIBUTION / MO	PAYBACK	CHURN	NET LTV	LTV:CAC	MOAT
A · App-only	$0	$9.99	$0	$7.44	—	7%	$61	2.4×	Weak
B · Free device + sub	$0	$14.99	$55	$11.54	4.8 mo	4%	$189	5.2×	Strong
C · Sell device + sub	$99	$9.99	$55 (+$44)	$6.69	day 1	4%	$166	4.7×	Strong
D · Tiered (blended)	$0	0 / 10 / 20	mix	blend	mixed	mixed	highest	best	Strong

Per member, blended, pilot-year costs. Contribution = revenue − processing − infra/AI/maps − support. Net LTV = lifetime contribution − device − CAC. Figures illustrative (Section 08 assumptions).

NET LIFETIME VALUE PER MEMBER, BY MODEL



3×

LIFETIME VALUE OF THE DEVICE MODEL VS APP-ONLY

4.8 → 2.5 mo

DEVICE PAYBACK, PILOT → VOLUME

5–11×

LTV:CAC AS THE FLYWHEEL COMPOUNDS

THE ONE-LINE READ

App-only earns ~$61 a member and can be cloned in a weekend. The free device earns ~$189 (up to ~$350 at scale) **and** builds a moat — but only if members stay long enough to pay back the ~$55 device. That is the entire game, and it lives or dies on churn.

06

THE DECISIVE RISK

The churn cliff.

This is the number that decides whether "free device" is brilliant or bankrupt. Give away a $55 device and you must keep the member long enough to earn it back. Below ~6% monthly churn the model is excellent; past ~8% it starts bleeding; past ~12% every member is a loss.

MONTHLY CHURN	AVG LIFE	NET LTV / MEMBER	VERDICT
3%	33 mo	+$281	Excellent
4%	25 mo	+$189	Strong
6%	17 mo	+$93	OK
8%	12 mo	+$44	Thin
12%	8 mo	−$4	Underwater
20%	5 mo	−$42	Loss

NET LTV PER MEMBER VS MONTHLY CHURN

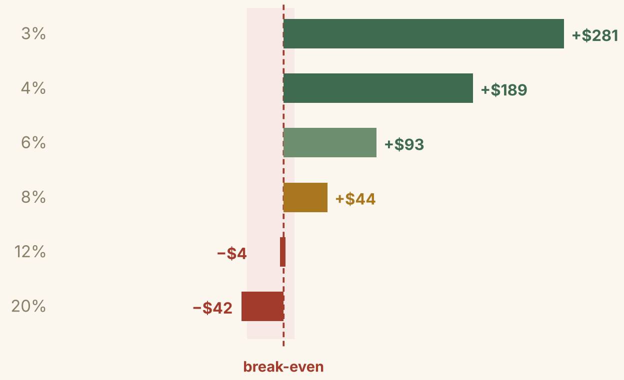


Fig. 3 — Free-device model, pilot costs. Left of the dashed line, every member loses money.

STAY LEFT OF THE CLIFF

Two levers, in order of power: **(1)** the 12-month commitment (structurally caps churn), and **(2)** genuine off-grid safety value people won't cancel before a season. Everything else — discounts, features, marketing — is secondary to these two.

07

THE AI EDGE

Stop predicting. Start rescuing.

Predicting terrain you've never seen on sand that moves is not a product — it's a liability. A safety tool that's confidently wrong is worse than none. So we kill the crystal ball and build the thing that's actually reliable and valuable: a reactive recovery co-pilot backed by data nobody else has.

WE DO NOT BUILD	WE DO BUILD
"You'll lose a vehicle over the ridge"	"Vehicle 3 is 1.2 km back and the gap is growing" (pure arithmetic on GPS — reliable)
Predicting hazards on unknown trails	Guided recovery when someone is stuck, from a curated expert playbook
Live photo diagnosis in the dunes	Queue the photo, analyse it in the cloud when reconnected — to improve the playbook

MEASURING ≠ PREDICTING

We keep the alerts that are just math on positions we already have. We drop anything that pretends to forecast the desert.

HOW THE RECOVERY CO-PILOT WORKS — AND WHY IT'S LIGHT

Tap "I'm stuck." Answer a few quick prompts — surface (sand / mud / rock), nose up or down, recovery gear on hand, solo or in a group, aired down? Get a step-by-step recovery procedure.

~90% is a deterministic expert playbook authored by real recovery pros. Near-zero compute, fully offline, and trustworthy *because it's curated, not hallucinated.* A small offline model on the **phone** just handles phrasing and follow-ups.

THIS PROTECTS THE ECONOMICS

The LoRa device runs **zero AI** — it's a dumb, cheap comms box. AI lives on the phone (offline) and in the cloud (when reconnected). Keeping the device dumb keeps the BOM low, which is exactly what makes the free device survivable.

The real AI moat isn't the model — it's **the knowledge base.** *Real stucks, real fixes, tagged by terrain and vehicle, compounding every trip. Anyone can copy a chat box; nobody can copy* **"here's what worked in Liwa in August."**

08

REALITY CHECK

The hidden cost of "free".

The free-device model is a working-capital business wearing a software costume. You pay for the device today and earn it back over months — a cash gap that grows with success. Respect it, and design around it.

CONSIDERATION	MITIGATION
~$55 out the door before month-5 payback	Annual plan billed up front funds the device on day one
Inventory & minimum order sizes tie up cash	Start with off-the-shelf Meshtastic-class units — no custom tooling
Hardware fails; returns cost real money	Reserve built into landed cost; simple, rugged, few parts
FCC/CE certification for custom hardware	Use pre-certified modules until volume justifies own design
Cancels before payback = guaranteed loss	12-month commitment; optional refundable device deposit

ASSUMPTIONS BEHIND EVERY NUMBER IN THIS BRIEF

DRIVER	PILOT	VOLUME
Landed device cost	$55	$32
Flagship subscription	$14.99	$14.99
Annual plan (effective/mo)	$12.42	$12.42
Infra + AI + maps / user	$2.00	$1.20
Support + ops / user	$1.00	$0.70
Payment processing	3%	3%
Blended CAC	$45	$35
Monthly churn	4%	3%

Illustrative planning assumptions — validate with suppliers and a live pilot before any capital commitment.

09

THE RECOMMENDATION

Option B economics, Option D packaging.

Don't sell the device (it strangles the growth that makes you defensible). Don't stay app-only (no moat). Sell tiers to capture the whole market, and steer serious groups to the free-device flagship on an annual commit — where the margin and the moat both live.

DO THIS

Prove willingness-to-pay first. Add hardware only once churn is proven.

Validate the app before you gamble on inventory. Then pilot the device small, watch the churn cliff, and scale only when the numbers hold.

PHASED ROADMAP

PHASE	OFFER	PRICE	GOAL	GATE TO NEXT
P1 · Validate	App-only Pro (maps, convoy-over-cell, trips)	$9.99/mo	Prove willingness-to-pay, build funnel — **zero hardware risk**	Healthy paid conversion & retention
P2 · Pilot device	Invite-only Convoy tier, off-the-shelf device	$149/yr	Measure real churn & device economics on a small fleet	Monthly churn < ~4%, payback confirmed
P3 · Scale	Open Convoy tier; negotiate BOM down	$149/yr	Grow mesh density & data moat; drive unit cost to volume	Volume economics & supply stable
P4 · Custom HW	Own-design device once demand is proven	bundled	Maximise margin, differentiation & IP	—

TIERS TO LAUNCH (P3+)

FREE

$0

Basic maps, single-vehicle trips, community trails. The funnel.

PRO

$9.99/mo

Offline maps, convoy-over-cell, recovery co-pilot, trip history. Solo overlanders.

CONVOY · FLAGSHIP

$149/yr

Free LoRa device, off-grid mesh tracking, SOS relay, priority support. Serious groups.

10

OVER TO YOU

The decisions in front of us.

THREE CALLS TO MAKE NOW

1. **Wedge.** Do we commit to **safety-first** (off-grid cohesion + recovery) as the headline, over social/discovery?
2. **Sequencing.** Ship **app-only Pro first** to validate, then pilot hardware — agreed?
3. **Price anchor.** Flagship at **$149/yr annual commit** (the churn lever), or do we also test a monthly?

WHAT WOULD MAKE THE MOAT REAL, FAST

The recovery knowledge base is the compounding asset. The fastest way to seed it is to **co-author the first playbooks with real desert-recovery pros or a club** — that authored expertise is both the seed data and instant credibility. If you have those relationships, that's move number one.

BOTTOM LINE

The app gets people in. The free device, the mesh, and the recovery data keep them — and keep everyone else out. Price for the system, not the software.

Illustrative planning model — not a forecast. All cost, churn, CAC and pricing figures are assumptions for comparison only and must be validated with suppliers and a live pilot before any capital commitment. Average member life approximated as 1 ÷ monthly churn. Contribution excludes fixed overhead (R&D, G&A) and one-time tooling/certification. Competitor pricing indicative and subject to change. Prepared for internal ideation · Overland · July 2026.